
Final Report

Introduction to Machine Learning Safety CARLA Autonomous Driving Perception System

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Contents

1	System Description	2
1.1	Provided system description	2
1.2	Key design limitations (by construction)	3
1.3	Known Design Limitation	3
1.4	Architecture & Training	3
2	Operational Design Domain	4
2.1	ODD Definition	4
2.2	ODD Gap Analysis	4
2.3	OOD Coverage Analysis	5
2.4	ODD Violation Response	5
3	Safety Analysis (STPA)	6
3.1	Control Structure	6
3.2	Losses	7
3.3	Hazards	7
3.4	Unsafe Control Actions (UCAs)	8
3.5	Causal Loss Scenarios	9
3.6	Safety Constraints	10
4	Safety Case: Verifications V-1 to V-5	11
4.1	Verification Summary	11
4.2	V-1: In-Distribution Recall	11
4.3	V-2: Adversarial Robustness	13
4.4	V-3: Calibrated Uncertainty	14
4.5	V-4: Out-of-Distribution Detection	15
4.6	V-5: Safe System Fallback	16
5	Residual Risks and Mitigations	18
5.1	Known Design Limitations :	19
6	Deployment Recommendation	20
A	Additional Material	23

1. System Description

Provided system description

An autonomous vehicle (AV) prototype operates in an urban environment. The system is composed of the following components:

1. **Camera:** A single forward-facing RGB camera, rigidly mounted behind the wind-shield, provides images at 10 Hz. There are *no other perception sensors* (no radar, no LiDAR, no ultrasonic).
2. **Perception models:** Three separate binary classifiers each receive the camera frame and output one binary prediction: *pedestrian present*, *vehicle present*, or *traffic light present*. Each model was trained exclusively on sunny daytime data collected in the CARLA simulator.
3. **Distance oracle:** A separate model estimates the distance to detected pedestrians and vehicles. You may assume this model is perfect for the sake of this course.
4. **Planning module:** The rule-based planner consumes the three perception outputs together with the current vehicle speed and steering angle. Its primary safety-relevant decisions are whether to continue driving, request deceleration, or issue an emergency brake command. Emergency braking is triggered when the pedestrian detector reports a pedestrian within a critical distance, when the vehicle detector indicates a road user ahead within a speed-dependent threshold, or when the traffic-light detector indicates that stopping is required at an approaching intersection. The planner does not have access to raw camera images and relies entirely on the perception models' outputs.
5. **Vehicle actuators:** Throttle, brake, and steering commands are executed by drive-by-wire actuators. Emergency braking applies maximum deceleration within the physical limits of the vehicle.
6. **Human safety operator:** A trained operator sits in the driver seat and monitors the road. The operator can override the autopilot at any time by pressing the brake pedal or turning the steering wheel. The operator receives a visual dashboard showing the current perception output and system status, but no auditory alerts are provided. During testing, operators work 4-hour shifts.
7. **Operating environment:** The vehicle is tested on public urban roads at speeds up to 50 km/h. The intended conditions are daytime, dry weather, and mapped intersections. However, weather and lighting conditions can change during a test drive (e.g. sudden cloud cover, low sun angle, rain onset).

Key design limitations (by construction)

- No sensor redundancy – a single camera is the only perception input.
- No other safety mechanisms are deployed yet.
- The human operator is the only fallback if the automation fails.
- The operator may exhibit a non-zero probability of delayed or missed intervention, especially under prolonged monitoring.

Known Design Limitation

The traffic-light detector reports only the presence of a traffic light, not its state (red / green). The planner therefore cannot, on its own, decide whether to stop at a signalized intersection. This is a limitation of the system as specified that no model metric can close, so it cannot be resolved within the safety analysis.

Architecture & Training

All three binary classifiers (**Pedestrian**, **Vehicle**, **Traffic light**) share an identical architecture and training system:

- **Backbone:** ResNet-18, pre-trained on ImageNet
- **Final layer:** Linear layer mapping 512 features to 2-class logits (Negative / Positive)
- **Optimizer:** Adam with learning rate $\eta = 1 \times 10^{-4}$ (Fine-tuning regime)
- **Batch size:** 32
- **Epochs:** 5 (fixed); best-validation-loss checkpoint saved, with a plateau-based learning-rate scheduler
- **Image preprocessing:** Resize to 224×224 , normalized with ImageNet mean/ Std
- **Loss function:** CrossEntropyLoss with class weights to handle imbalance
- **Dataset:** 7,200 training frames, 3,600 validation frames, 3,600 test frames are all captured in CARLA under sunny daytime conditions.

Class Distribution: Training Set:

Label	Present (True)	Absent (False)
Traffic Light	5,276 (73.3%)	1,924 (26.7%)
Pedestrian	1,718 (23.9%)	5,482 (76.1%)
Vehicle	5,458 (75.8%)	1,742 (24.2%)

Table 1: *Training dataset class distribution*

The pedestrian class is considerably under-represented (**23.9%**), which contributes to the severe overfitting and low recall observed during training.

2. Operational Design Domain

The system is designed to operate under specific environmental and operational conditions. This section defines the intended ODD, identifies gaps between the ODD and training data, and quantifies how well the test set samples the ODD.

ODD Definition

ODD Dimension	Operating Conditions	Non-Operating Conditions	Detection
Lighting	Daytime (sun elevation angle $\geq 15^\circ$)	Dusk / night (sun elevation $< 15^\circ$)	Yes – MSP OOD monitor (traffic-light model), AUROC 0.537 (weak)
Weather	Dry, clear sky	Rain, fog, snow	Yes – MSP OOD monitor (traffic-light model), AUROC 0.737
Scene type	Mapped urban roads (Training town layout)	Unmapped rural / off-road / unseen town layouts (Town-01)	Yes – MSP OOD monitor (traffic-light model), AUROC 0.812
Vehicle speed	Continuous operation ≤ 50 km/h	Speeds > 50 km/h	Not evaluated – no OOD detector tested for this dimension
Road geometry	Straight segments and gentle curves ($R > 100$ m)	Steep grades, tight radius curves	Not evaluated – no OOD detector tested for this dimension
Camera state	Normal operation (no occlusion)	Dirty lens, rain drops, condensation	Not evaluated – no OOD detector tested for this dimension

ODD Gap Analysis

Two important ODD dimensions have zero training coverage as training data covers only sunny daytime conditions. :

1. **Weather:** System is intended to operate in dry conditions, but OOD test includes fog scenarios. There is no foggy training data available.
2. **Scene:** System is intended to operate only on urban road layouts present in the training dataset, but the OOD test set includes Town-01 (an unseen geographic architecture)

OOD Coverage Analysis

Evidence (k -projection coverage):

k	Coverage in Test Set
1	55.56% : Only 5 out of 9 individual ODD dimensions are well-represented in test set. Specifically, some combinations of weather (rainy), lighting (night), and speed (high) are undersampled.
2	25.93% : Only 7 out of 27 pairwise ODD combinations are covered. Many two-dimensional combinations are absent (e.g., rainy + night, high speed + dusk).
3	11.11% : Only 3 out of 27 three-way combinations are covered. Higher-order interactions are largely missing from test set.

Interpretation: As k increases, coverage degrades sharply ($k = 1$: 55.56%, missing clear weather, dusk/dawn; $k = 2$: 25.93%; $k = 3$: 11.11%). Verification evidence is valid only for the **sunny-daytime-urban** ODD subset; uncovered conditions carry to residual risk. Conditions which are left uncovered are carried to the residual risk in Section 5.

ODD Violation Response

When an out-of-domain condition or anomalous visual input (such as heavy fog, nighttime lighting, or an unmapped town layout) is detected at runtime, the autonomous driving architecture initiates the following graded fallback sequence:

1. **Confidence suppression:** Flagged perception outputs are treated as untrustworthy, and confidence values are suppressed to prevent the planner from executing standard cruising or acceleration commands.
2. **Deceleration request:** The rule-based planner commands a controlled deceleration profile to bring the vehicle to a safe stop within the kinematic limits of the roadway.
3. **Operator alert:** The visual dashboard immediately alerts the human safety operator to execute a manual takeover within the 1.5-second reaction budget.
4. **Manual override opportunity:** Physical inputs to the brake pedal or steering wheel immediately override autopilot actuation via the drive-by-wire system.

3. Safety Analysis (STPA)

Control Structure

The following control structure captures all sensors, feedback paths and the human operator. The planning module acts on the vehicle only through the binary labels produced by the three models, it has no direct access to raw camera images.

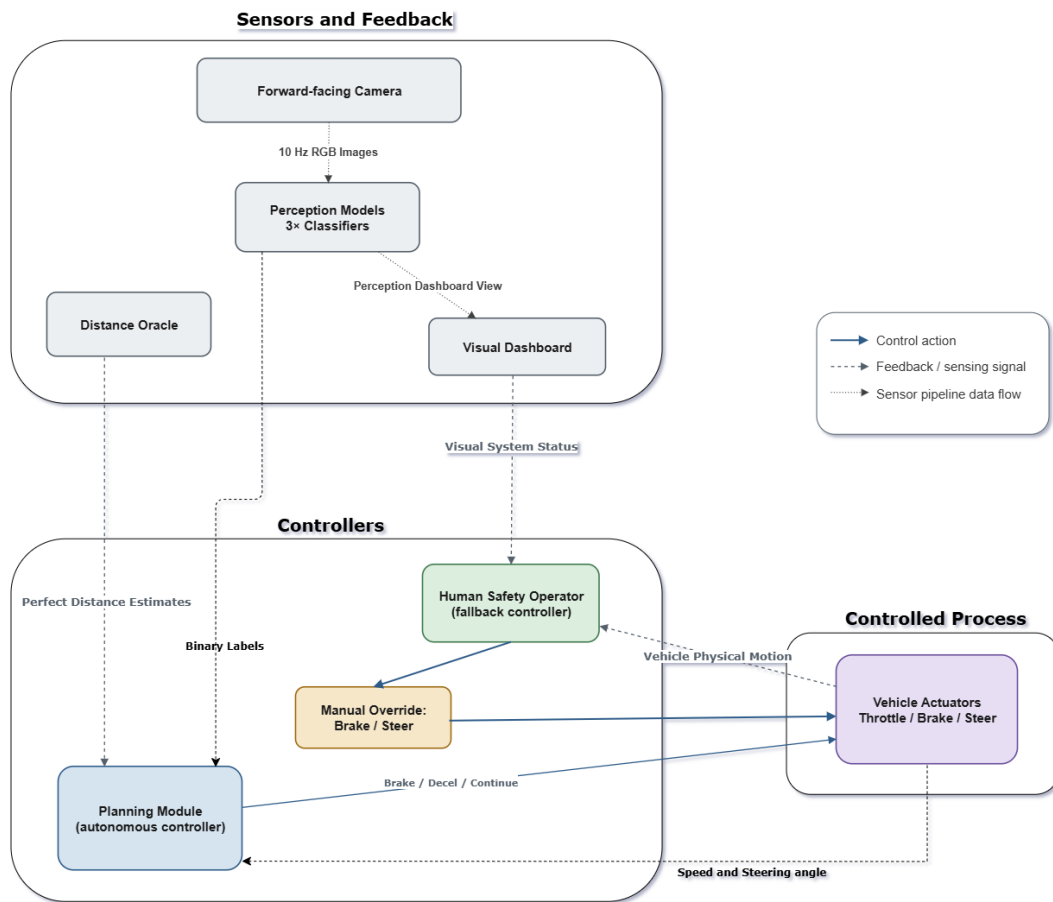


Figure 1: Complete **STPA** control structure of the CARLA perception system.

Losses

ID	Loss	Why Unacceptable ?
L-1	Injury or loss of human life.	Primary ethical and moral imperative of safety-critical systems.
L-2	Material damage (AV or property).	High financial repair costs and liability.
L-3	Violation of traffic laws/fines.	Leads to legal liability and potential loss of testing license.
L-4	Loss of public trust/Reputation.	Social and ethical backlash can end/delay the entire AV program.

Hazards

ID	Hazard	Loss(es)
H-1	Pedestrian in path while vehicle is moving.	L-1, L-2
H-2	Autonomous mode remains active outside the ODD, or operates on undetected out-of-distribution (OOD) visual inputs, leading to untrustworthy perception outputs	L-1, L-2, L-3
H-3	Approaching intersection without detecting stop condition.	L-1, L-2, L-3
H-4	Planner commands acceleration when obstacle is in path.	L-1, L-2, L-4
H-5	Autonomous mode operates on undetected, adversarially perturbed visual inputs, leading to corrupted, high-confidence false perception outputs.	L-1, L-2, L-3

Unsafe Control Actions (UCAs)

ID	Controller	Control action	UCA type	Hazards	Context in which unsafe
UCA-1	Planner	Continue Driving	Provided unsafely	H-1	Pedestrian detected in path but vehicle does not stop.
UCA-2	Planner	Emergency Brake	Not provided	H-1	Pedestrian within critical distance but brake not issued.
UCA-3	Planner	Emergency Brake	Wrong timing	H-3	Braking issued too late to stop at a detected intersection.
UCA-4	Operator	Manual Override	Not provided	H-2	Automation fails outside ODD; operator fails to take over.
UCA-5	Planner	Emergency Brake	Provided unsafely	H-4	Hard brake issued on highway with no obstacle, causing rear-end.
UCA-6	Planner	Request Decel	Not provided	H-3	Traffic light detected (presence) but no slowdown initiated.
UCA-7	Operator	Manual Override	Wrong timing	H-1	Operator reacts too late (> 1.5 s) to a missed ML detection.
UCA-8	Planner	Continue Driving	Wrong timing	H-3	Continues through intersection after light presence was detected.
UCA-9	Planner	Continue Driving	Provided unsafely	H-2, H-1, H-3	The planner continues driving at nominal speed while camera input is out-of-distribution (e.g., heavy fog or night) and perception outputs are untrustworthy because distribution shift went undetected.
UCA-10	Planner	Continue Driving / Accelerate	Provided unsafely	H-5, H-1, H-3, H-4	The planner continues driving or commands acceleration because a successful adversarial attack has silently hidden an obstacle (false negative), causing the system to act as if the path is clear.

Causal Loss Scenarios

LS-ID	For UCA	Causal scenario (why the UCA occurs)
LS-1	UCA-1,2,7,9	Low recall in-distribution: Under the (sunny daytime) conditions, visual challenges like small target scale or severe class imbalance causes the classifier to miss an obstacle (false negative). The model gives output “absent”, so planner receives no alert and continues at speed rather than utilising emergency brakes.
LS-2	UCA-1, UCA-10	Adversarial perturbation: Subtle, worst-case input noise (such as gradient-based perturbations or physical adversarial artifacts on objects) shifts feature representations across the decision boundary. The detector outputs a false negative with high certainty, which prevents the planner from detecting the hazard ahead.
LS-3	UCA-1, UCA-2	Overconfident false negative bypasses fallback: The pedestrian classifier produces a false negative – a pedestrian is present but the model outputs “absent” – while simultaneously reporting high confidence in that wrong prediction (avg. confidence on incorrect predictions = 0.7387, Exercise 9.4). Because the planner treats high confidence as a reliability signal, it registers “no pedestrian, reliable” rather than triggering any low-confidence fallback, allowing the incorrect perception label to propagate silently.
LS-4	UCA-9, UCA-10	Undetected out-of-distribution input: Environmental conditions outside the training domain (such as fog, nighttime lighting or unfamiliar town) distort the feature representations. The OOD monitor fails to detect the domain shift, leading the planner to act on corrupted perception outputs under the false assumption of normal operating conditions.
LS-5	All UCA types	Fallback and takeover breakdown: A model-level error propagates through the perception pipeline without triggering an automated minimum risk maneuver. Due to monitoring fatigue, absence of auditory alerts, or a takeover reaction time exceeding safe kinematic stopping margins (> 1.5 s), the human operator fails to execute manual override in time.

Safety Constraints

SC-ID	Controls (LS-ID)	Safety constraint (requirement)	Level	Verified by
SC-1	LS-1, LS-2, LS-3, LS-5	Reliably detect pedestrians in the vehicle's path; do not continue driving once a pedestrian is detected within the critical distance.	Model	V-1, V-3
SC-2	LS-1, LS-3, LS-5	Pedestrian within the critical distance detected with recall high enough to trigger emergency braking.	Model	V-1, V-3
SC-3	LS-5	Traffic lights detected reliably and early enough for the planner to brake before the stopping point.	Model	V-1
SC-4	LS-5	Outside the defined ODD, autonomous operation shall be disabled or the operator explicitly required to take control.	System	—
SC-5	LS-5	False-positive obstacle detections minimized so emergency braking is not triggered without a relevant obstacle.	Model	—
SC-6	LS-5	Mere traffic-light presence not treated as sufficient to stop; a valid light state must be available first.	System	—
SC-7	LS-1, LS-5	Sufficient warning and monitoring for the safety operator to intervene before a missed detection becomes hazardous.	System	—
SC-8	LS-5	No continuing through an approaching intersection unless available traffic information establishes it is safe.	System	—
SC-9	LS-1, LS-4, LS-5	OOD monitor reliably discriminates out-of-distribution inputs (fog, night, unseen towns) from in-distribution ones; detector choice (MSP, Mahalanobis, k -NN) justified by measured performance.	Model	V-4
SC-10	LS-4, LS-5	On an OOD or ODD-violation flag, inhibit driving commands and prompt operator takeover within 1.5 s, or auto-execute a Minimum Risk Maneuver if unresponsive.	System	V-5
SC-11	LS-2, LS-4, LS-5	Recall drop under adversarial perturbation ($\epsilon \in \{0.01, 0.05, 0.1\}$) empirically measured for all three classifiers and minimized via defensive measures.	Model	V-2
SC-12	LS-5	On a flagged adversarial input or broken confidence calibration, inhibit speed/acceleration commands and execute a Minimum Risk Maneuver within 1.5 s, alerting the operator.	System	V-5

4. Safety Case: Verifications V-1 to V-5

This section presents evidence that each safety constraint is met, partially met, or not met. Each verification is tied to a loss scenario and ultimately to a loss.

Verification Summary

Verification	Verifies SC(s)	Loss scenarios	Hazards
V-1: In-distribution Detection Performance	SC-1, SC-2, SC-3	LS-1	H-1, H-2, H-3
V-2: Robustness to perturbations	SC-11	LS-2	H-1, H-2
V-3: Calibrated uncertainty	SC-1, SC-2 (via LS-3)	LS-3	H-1, H-2, H-4
V-4: Out-of-distribution detection	SC-9	LS-4	H-1, H-2, H-3, H-5
V-5: Safe system fallback	SC-10, SC-12	LS-5	All Hazards

V-1: In-Distribution Recall

Verifies: SC-1, SC-2, SC-3 **Hazard(s):** H-1, H-2, H-3 **Loss scenario(s):** LS-1
Constraint: In-distribution test recall meets target safety thresholds under nominal ODD conditions: pedestrian ≥ 0.90 , vehicle ≥ 0.85 , and traffic light ≥ 0.85 .
Metric: Per-class recall on the in-distribution test set.

Evidence (In-Distribution Detection Performance) — Exercise 3

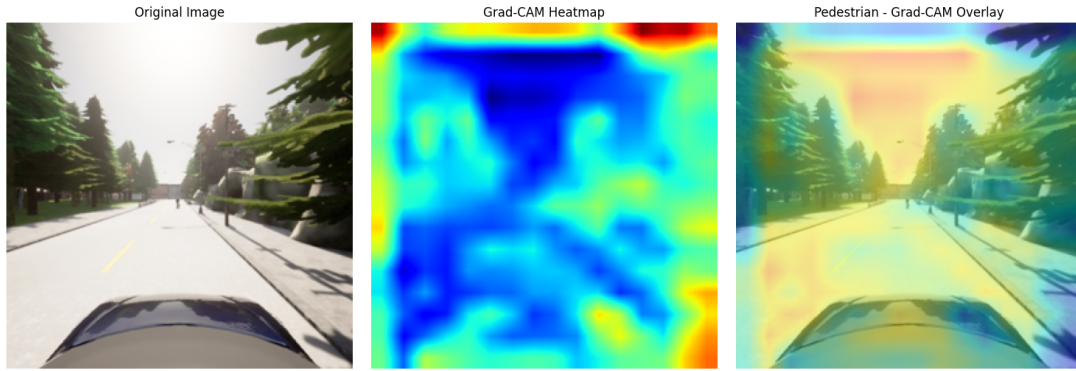
Model	Recall	Threshold	Met?
Pedestrian	0.483	≥ 0.90	✗ Not met
Traffic light	0.983	≥ 0.85	✓ Met
Vehicle	0.848	≥ 0.85	✗ Not met

Overall verdict: ✗ **Not met**

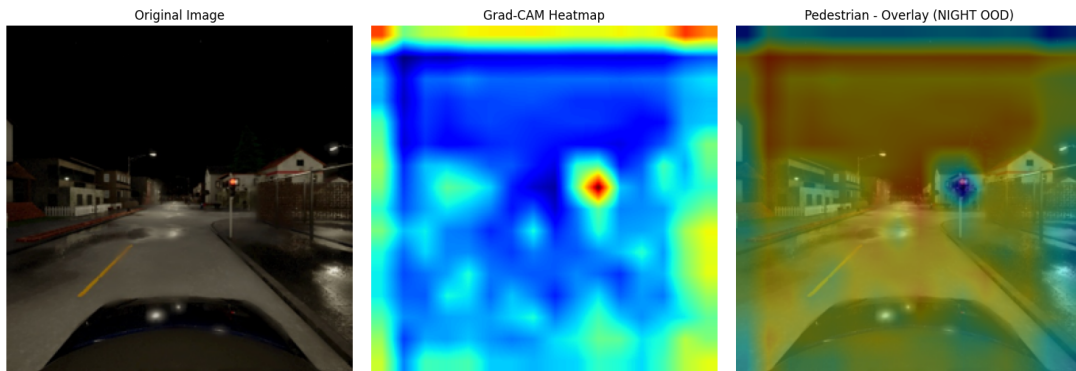
Interpretation:

- **Pedestrian:** Recall ≈ 0.483 *does NOT meet* threshold of 0.90. The detector misses approximately 52% of pedestrians in the test set.
Primary cause: *severe class imbalance and overfitting identified in Exercise 3.5.*
- **Traffic Light:** Recall ≈ 0.983 *exceeds* threshold of 0.85. Good performance on this well-defined task.
- **Vehicle:** Recall ≈ 0.848 *does NOT meet* threshold of 0.85. Falls just short despite otherwise strong precision — primary cause is the 411 false negatives (missed vehicles) out of 2,700 positive test cases.

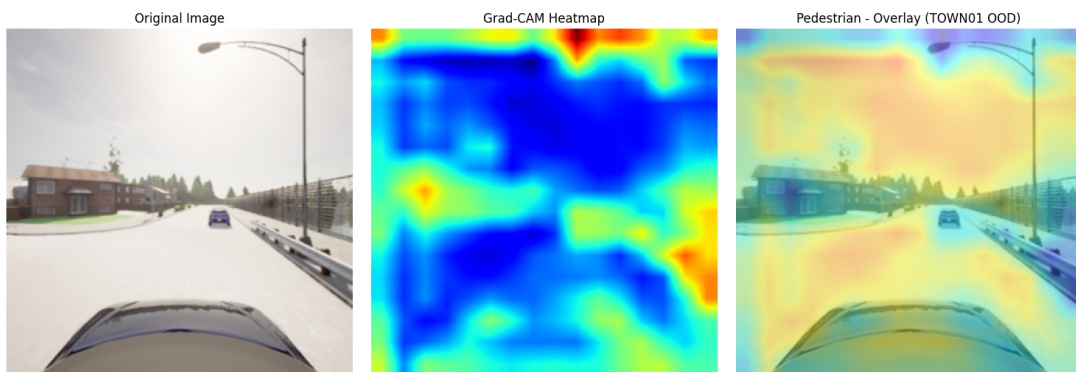
Supporting evidence from Grad-CAM (Exercise 6): Explainability analysis (Exercise 6.5, Task 3) gives a mixed result, not a uniform pass. Some heatmaps correctly localize on the pedestrian (Fig. 2a), but others attend to background regions – sky, tree-line, or building edges – rather than the target object, even in frames with no vehicle present. This helps explain the recall shortfall in V1-evidence table: the model’s attention is not reliably anchored to the object it is meant to detect. Traffic-light heatmap findings are inconclusive, since the sampled images did not consistently contain a visible traffic light.



(a) In-distribution (daytime)



(b) Nighttime (OOD)



(c) Different CARLA town (OOD)

Figure 2: Grad-CAM overlays for the pedestrian detector across 3 conditions.

V-2: Adversarial Robustness**Verifies:** SC-11 **Hazard(s):** H-1, H-2 **Loss scenario(s):** LS-2**Constraint:** Recall does not degrade substantially under small adversarial perturbations, limiting the impact of adversarial manipulation of the camera feed.**Metric:** Recall drop under FGSM at $\varepsilon = 0.05$ (clean $\rightarrow$ adversarial).**Evidence (Adversarial Robustness) — Exercise 8**

Model	Recall (clean)	Recall $\varepsilon = 0.05$	Drop	Met?
Pedestrian	0.52	0.09	83%	× Not met
Traffic light	0.99	0.47	53%	× Not met
Vehicle	0.83	0.54	35%	× Not met

Overall verdict: × **Not met**

Note: Evaluated on a 100-image stratified subset of the in-distribution test set (Exercise 8.5), not the full 3,600-image split.

Root cause: Standard deep neural networks are known to be vulnerable to adversarial examples. ResNet-18, while accurate on natural images, offers no built-in defense against adversarial perturbations. Defenses would require retraining models, which is out of scope, and this is why **SC-11** is not met.

V-3: Calibrated Uncertainty

Verifies: SC-1, SC-2 (via LS-3) **Hazard(s):** H-1, H-2, H-4 **Loss scenario(s):** LS-3

Constraint: The models provide well-calibrated confidence estimates, so low-confidence predictions can trigger fallback behaviour rather than silently overcommitting.

Metric: Expected Calibration Error (ECE) on the in-distribution test set, before and after temperature scaling.

Evidence (Uncertainty) — Exercise 9

Model	ECE (single)	ECE (temp. scaled)	Met?
Pedestrian	0.1150	0.0331 (T=2.10)	✓ Met
Traffic light	0.0330	0.0275 (T=1.20)	✓ Met
Vehicle	0.0459	0.0248 (T=1.30)	✓ Met

Overall verdict: ✓ Met

***Exercise 9.4** measures raw ECE and **Exercise 9.5** applies temperature scaling and measures improved ECE.*

*Data extracted from **Exercise 9.5 JSON** results file. All detectors are significantly overconfident before scaling (high ECE); temperature scaling reduces ECE substantially.*

Interpretation:

- All three detectors achieve $ECE < 0.05$ after temperature scaling.
- The pedestrian detector is severely overconfident in raw form ($ECE = 0.1150$), reflecting misalignment between confidence and accuracy. Temperature scaling by $T = 2.10$ corrects this.
- Scaled models are well-calibrated: pedestrian confidence now reflects accuracy within 3.31% error, traffic light within 2.75%, vehicle within 2.48%.

Operationalization:

1. Apply temperature scaling to logits before computing confidence.
2. Store optimal temperatures ($T_{\text{pedestrian}} = 2.10$, $T_{\text{traffic light}} = 1.20$, $T_{\text{vehicle}} = 1.30$) in config files.
3. Feed the calibrated probability directly into the cost-optimal decision rule (SC-10, V-5, $\tau^* \approx 0.0099$) rather than gating on a fixed confidence threshold – calibration ensures this probability is trustworthy, while the threshold itself is set by asymmetric cost, not by confidence alone.

V-4: Out-of-Distribution Detection

Verifies: SC-9 **Hazard(s):** H-1, H-2, H-3, H-5 **Loss scenario(s):** LS-4

Constraint: An accompanying monitor reliably flags inputs that fall outside the ODD, preventing silent failure on conditions not covered by training data (fog, night).

Metric: AUROC of the MSP-based OOD monitor, evaluated on the traffic-light classifier, separating in-distribution from OOD inputs.

Evidence (Anomaly / OOD Detection) — Exercise 7

Detector used: Maximum Softmax Probability (MSP) – lower MSP scores tend to indicate higher OOD possibility.

Model	Detector	OOD set	AUROC	Met?
Traffic light	MSP	Town-01	0.812	≈ Partial
Traffic light	MSP	Fog	0.737	× Not met
Traffic light	MSP	Night	0.537	× Not met

Overall verdict: ≈ **Partial**.

*The MSP detector falls well short of the $AUROC \geq 0.90$ threshold across all OOD splits, with night-time input (0.537) barely separable from in-distribution – the model remains confidently wrong precisely when scene appearance changes most. Town-01 (unseen map, same weather) is comparatively more separable (0.812) but still non-compliant. SC-9 is **not met** under the MSP baseline.*

Scope note: OOD-AUROC was evaluated only for the traffic-light detector (Exercise 7.6/7.7). No equivalent OOD-detection evidence exists for the pedestrian or vehicle classifiers – this verification does not establish that OOD monitoring covers H-1 (pedestrian) or the vehicle-detection failure mode, despite both being listed as hazards addressed by SC-9.

V-5: Safe System Fallback

Verifies: SC-10, SC-12 **Hazard(s):** All Hazards **Loss scenario(s):** LS-5

Constraint: the system provides a safe fallback when a perception model fails or operates outside the ODD - even if a model produces incorrect output, the architecture limits propagation of the error to harm through fallback behaviour and human oversight.

Evidence (Design Argument) — Exercise 9.6**Fallback behaviour (By design):**

- **Cost-optimal trigger:** the planner requests deceleration when the pedestrian-detector output probability exceeds $\tau^* = C_{FN}/(C_{FN}+C_{FP}) = 100/101 \approx 0.0099$, using asymmetric costs $C_{FN} = 100$ (missed pedestrian - catastrophic loss, severe injury, major liability), $C_{FP} = 1$ (false alarm - minor inconvenience).
- **OOD trigger:** if the MSP-based monitor from (**Exercise 7.6**) flags the frame as out-of-distribution, the planner requests deceleration and the dashboard shows a low-confidence alert.
- **Human override option:** the operator can brake or take back control of the AV at any time.

System-level constraints (SC-10) addressed by design: lowering the decision threshold from $\tau = 0.5$ to $\tau^* \approx 0.0099$ on the pedestrian detector reduces false negatives from 365 to 0 (calibrated model) and total cost-weighted loss from 37,385 to 2,894 - at the expense of false positives rising from 885 to 2,894 (accuracy falls from 0.653 to 0.196). This trade is intentional: $C_{FN} \gg C_{FP}$ means the architecture accepts far more false alarms to drive missed-pedestrian events toward zero.

Standard vs. Cost-Optimal Autopilot Comparison :

Metric	Standard Autopilot ($\tau = 0.5$)	Cost-Optimal Autopilot ($\tau^* \approx 0.01$)	What Changed in Reality ?
Missed Pedestrians (False Negatives)	365	0	The car did not miss a single pedestrian across the test set.
False Alarms (False Positives)	885	2,894	Frequent phantom braking. The car slows down for shadows, signs, or empty road far more often.
Standard Accuracy	65.3%	19.6%	Raw classification accuracy collapses because the model guesses “pedestrian” almost constantly.
Total Risk / Cost-Weighted Loss	37,385	2,894	92.3% reduction in risk. Missing people is so costly that eliminating misses saves the system overall.

Limitations of the fallback:

- **OOD detector coverage:** The fallback’s OOD trigger relies on the MSP monitor from V-4, which does not reliably detect night-time OOD input AUROC (0.537) and is only partially reliable on fog (0.737) and unmapped town (0.812) – all below the **0.90** target.
- **No independent sensor:** The fallback trigger depends entirely on the same camera pipeline that may already be compromised (fog, night, adversarial input); there is no independent procedure to cross-check.
- **Human reaction time:** Even with the lowered threshold, response is still dependent on operator’s attention and vehicle braking distance, neither of which is modelled here.

Overall verdict: $\approx$ Partial

*Cost-optimal thresholding done in (**Exercise 9.6**) is effective and verified: False negatives on the pedestrian detector go from 365 to 0 at $\tau^* \approx 0.0099$. However, Safety Constraint (SC-10) depends on the OOD monitor (V-4) to trigger reliably and that monitor’s MSP baseline is well below the 0.90 target, especially at night (0.537). V-5 has been **partially met**: the cost-optimal decision rule is sound and verified statistically, but the fallback’s OOD trigger inherits the reliability gap which is already identified in V-4.*

5. Residual Risks and Mitigations

From	Verdict	Residual risk	Hazards	Possible mitigations
V-1	× Not met	The pedestrian detector’s recall (<i>0.483</i>) leaves roughly half of all pedestrians undetected; vehicle recall (<i>0.848</i>) also falls just short of its 0.85 threshold. Only traffic-light recall (<i>0.983</i>) clears its threshold. H-1 is inadequately controlled at the model level for both pedestrian and vehicle detection.	H-1, H-2	Rebalancing the pedestrian training set (currently 23.9% positive, 1,718 examples) via oversampling or focal loss to reduce overfitting (validation loss rose 56.1% while training loss fell 86.7%) through stronger regulation or early stopping.
V-2	× Not met	An FGSM perturbation of $\varepsilon = 0.05$, designed to be visually imperceptible, drops pedestrian recall by 83%, traffic-light recall by 53%, and vehicle recall by 35%. The camera pipeline has no built-in resistance to this kind of attack.	H-1, H-2, H-3	Introducing adversarial training (FGSM/PGD) examples in the training loop and adding input pre-processing such as JPEG re-compression or bit-depth reduction to evaluate whether a non-visual sensing modality can cross-check camera-only detections.
V-4	≈ Partial	The MSP-based OOD monitor (traffic-light model only) reaches AUROC (<i>0.812</i>) on the unseen town but only (<i>0.737</i>) on fog and (<i>0.537</i>) at night – effectively no better than chance in the dark. Nighttime OOD frames pass through mostly unflagged, and H-1/H-5 coverage beyond this one model remains unverified.	H-1, H-2, H-3, H-5	Replacing the MSP baseline with the feature-based detectors already evaluated (Mahalanobis distance reached 0.949 overall, k-NN 0.931) and restricting autonomous operation to daytime ODD until a night-capable detector is validated.
V-5	≈ Partial	Cost-optimal thresholding is itself verified (false negatives $365 \rightarrow 0$ on the pedestrian detector), but the fallback only fires when the OOD monitor or low-confidence signal triggers it. Because the monitor is unreliable at night, the fallback inherits V-4’s blind spot rather than closing it.	All Hazards	By not treating the OOD monitor as the only trigger and adding a confidence-independent fallback condition (scheduled deceleration in low-light ODD) and require operator acknowledgment rather than a passive dashboard alert.

Known Design Limitations :

These are architecture-level constraints and do not belong to the residual risk analysis (*which is for unmet safety constraints*) rather, they are limitations of the system design itself.

- **Traffic-light state is not detected (Leaves H-3 open irrespective of V-1–V-5 verdicts) :**
The traffic-light model reports only *presence*, not colour (red/green). The planner cannot decide whether a stop is legally required at a signalised intersection from this signal alone (Section 1). No amount of retraining the current binary classifier closes this gap it would require a different model output space entirely.
- **Single camera, no sensor redundancy (Leaves H-1, H-2, H-3 open; this is why V-2 and V-4 cannot be fully closed by better training alone) :**
All perception – pedestrian, vehicle, and traffic-light detection – depends on one forward-facing RGB camera. There is no radar, LiDAR, or ultrasonic backup. Any failure mode that degrades the camera feed (adversarial perturbation, fog, night, glare, physical occlusion) degrades *all three* detectors simultaneously, since they share the same single point of failure.
- **No auditory operator alert (Compounds the residual risk already identified for V-5):**
The human safety operator receives only a visual dashboard indicating perception output and system status, there is no auditory or haptic alert. During sustained visual monitoring this increases the chance a fallback trigger is missed, independent of how reliable the underlying OOD monitor is. Even a correctly firing fallback depends on the operator noticing a passive visual cue.
- **Operator fatigue over 4-hour shifts (Leaves H-1–H-3 open as a residual human factor behind every fallback path) :**
Operators are scheduled for 4-hour monitoring shifts (Section 1). Sustained low-intervention monitoring is a known human-factors risk for delayed reaction; this is a scheduling/procedural limitation, not something any perception model can fix.
- **Distance oracle assumed perfect (Leaves H-1, H-2 open as an unverified assumption underlying the entire planner) :**
The planner’s braking-distance decisions rely on a separate distance-estimation model that this course explicitly instructs to be treated as perfect (Section 1). No verification in this report tests that assumption. If the distance oracle is imperfect in deployment, the emergency-braking thresholds computed from it would be wrong regardless of how well the pedestrian/vehicle/traffic-light detectors perform.
- **Planner has no access to raw camera images (Leaves H-1–H-5 open as an architectural ceiling on top of all model-level verifications) :**
The rule-based planner consumes only the three binary perception outputs, never the underlying image (Section 1). It therefore cannot apply any independent sanity check (e.g. noticing a scene looks different) beyond what the perception models already report - it can only be as safe as the weakest upstream detector.

6. Deployment Recommendation

Recommendation: ☐ Deploy ☒ Deploy with restrictions ☐ Do not deploy

Conditions / restrictions:

- **Daytime operation only:** The OOD monitor is only weakly separable at night (AUROC 0.537, V-4); autonomous mode must stay off after dusk until a night-capable detector is validated.
- **Cost-optimal pedestrian threshold mandatory:** The planner must use $\tau^* \approx 0.0099$, not $\tau = 0.5$, for the pedestrian detector (V-5) - the standard threshold misses 365 pedestrians in the test set; the cost-optimal one drives this to zero.
- **Closed or controlled test routes only:** Public deployment is unsupported while V-2 (*Adversarial robustness*) is not met - an imperceptible $\varepsilon = 0.05$ perturbation drops pedestrian recall by 83% (0.52→0.09).
- **Mandatory auditory alert before public trials:** The dashboard is visual-only (Section 1); an auditory or haptic alert is needed before public-road shifts, since V-5's fallback depends on operator attentiveness to a passive cue.
- **Operator shift length should be capped below four hours:** Due to the known risk of monitoring fatigue.
- **No reliance on traffic-light colour:** The model detects presence only, not state; intersections stay human-supervised or externally signaled until a state-aware model replaces it.

Justification:

- **V-1 (× Not met):** Pedestrian recall (0.483) and vehicle recall (0.848) both miss their 0.90/0.85 thresholds - roughly half of pedestrians go undetected. Only traffic-light recall (0.983) clears its threshold. H-1 is inadequately controlled at the model level.
- **V-2 (× Not met):** All three detectors lose recall under a small FGSM perturbation - pedestrian worst at 83% (0.52 to 0.09), traffic-light 53%, vehicle 35%. This is an unmitigated attack surface on the single camera feed, not a marginal shortfall.
- **V-3 (✓ Met):** Every detector reaches ECE < 0.05 after temperature scaling, so the fallback's confidence signal can be trusted.
- **V-4 (≈ Partial):** The deployed MSP monitor reaches AUROC 0.812 on an unseen town but only 0.737 on fog and 0.537 at night - close to chance after dark. A feature-based alternative (Mahalanobis, 0.949 overall) performs far better but is not yet deployed.
- **V-5 (≈ Partial):** The cost-optimal fallback rule is verified and highly effective once triggered, but only as reliable as the OOD monitor triggering it - it inherits V-4's night-time blind spot rather than closing it.

Summary : *Only one of the five constraints **V-3** is fully met; **V-4** and **V-5** are partially met, while **V-1** and **V-2** - the latter an active, unmitigated adversarial vulnerability on the system's only sensor - are not met at all. An unrestricted deployment recommendation is therefore not defensible on the current evidence.*

*However, **V-5's** cost-optimal decision logic is quantitatively strong when it fires, and the calibration underlying it (**V-3**) is fully verified, so a restricted deployment - daytime, closed-route, cost-optimal threshold enforced, auditory alerting added - is justified for continued controlled testing.*

*The restrictions above should be lifted only once **V-1** (pedestrian recall ≥ 0.90), **V-2** (FGSM recall drop $< 10\%$), and **V-4** (AUROC ≥ 0.90 , particularly at night) are met in a future system revision.*

GitHub Repository Link:

<https://github.com/AbS2693/ITMLS--Abhishek-Bangale>

Statement of Authorship

I hereby declare that I have independently written the present work and have not used any sources or aids other than those stated.

Texts, trains of thought, concepts, graphics, etc. from external sources that I have directly or indirectly adopted in my work I have clearly marked as such and provided with complete references to the respective source. All other content of this work (text parts, figures, tables, etc.) without corresponding references originates in the copyright sense from me.

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I confirm that I have fully understood the content of my work and can represent it independently.

Magdeburg, 22.08.2026

Place, Date

Abhishek Bangale

Signature

A. Additional Material

Supporting Figures for Verifications V-1 to V-5 :

The figures below provide supporting visual evidence for claims made in Section 4 that are otherwise stated only as numbers or descriptions in the main text.

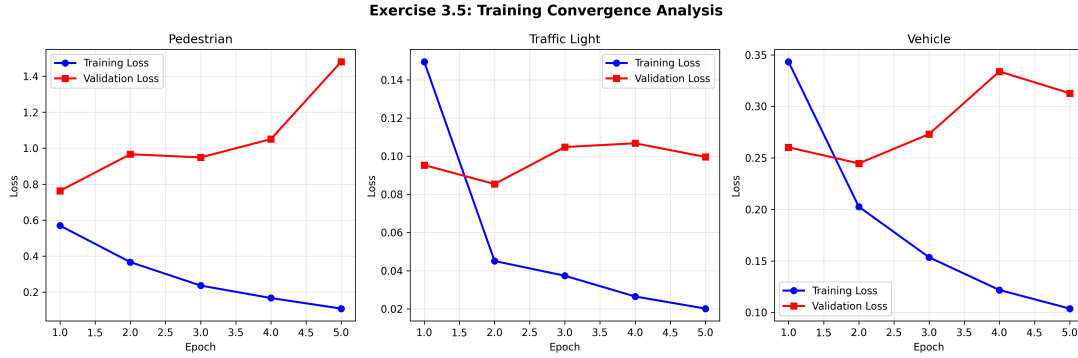


Figure 3: **Training vs. validation** loss curves for the pedestrian detector (**Exercise 3.5**). Validation loss rises while training loss falls, visually confirming the overfitting cited as the root cause of **V-1**'s pedestrian recall shortfall (**0.483 vs. 0.90** threshold).

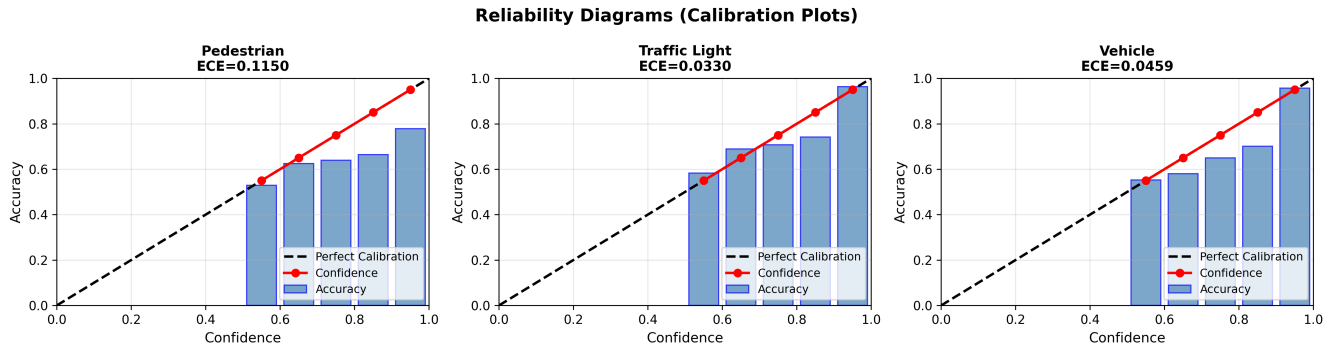


Figure 4: Reliability diagrams showing **confidence vs. accuracy** before temperature scaling (**Exercise 9.4**), visualising the overconfidence that **V-3**'s calibration constraint is designed to correct.



Figure 5: **Clean vs. FGSM-perturbed** input at $\epsilon = 0.05$ for the pedestrian detector (**Exercise 8.4**). The perturbation is visually imperceptible, supporting the **V-2** claim that this attack requires no physical tampering with the scene.

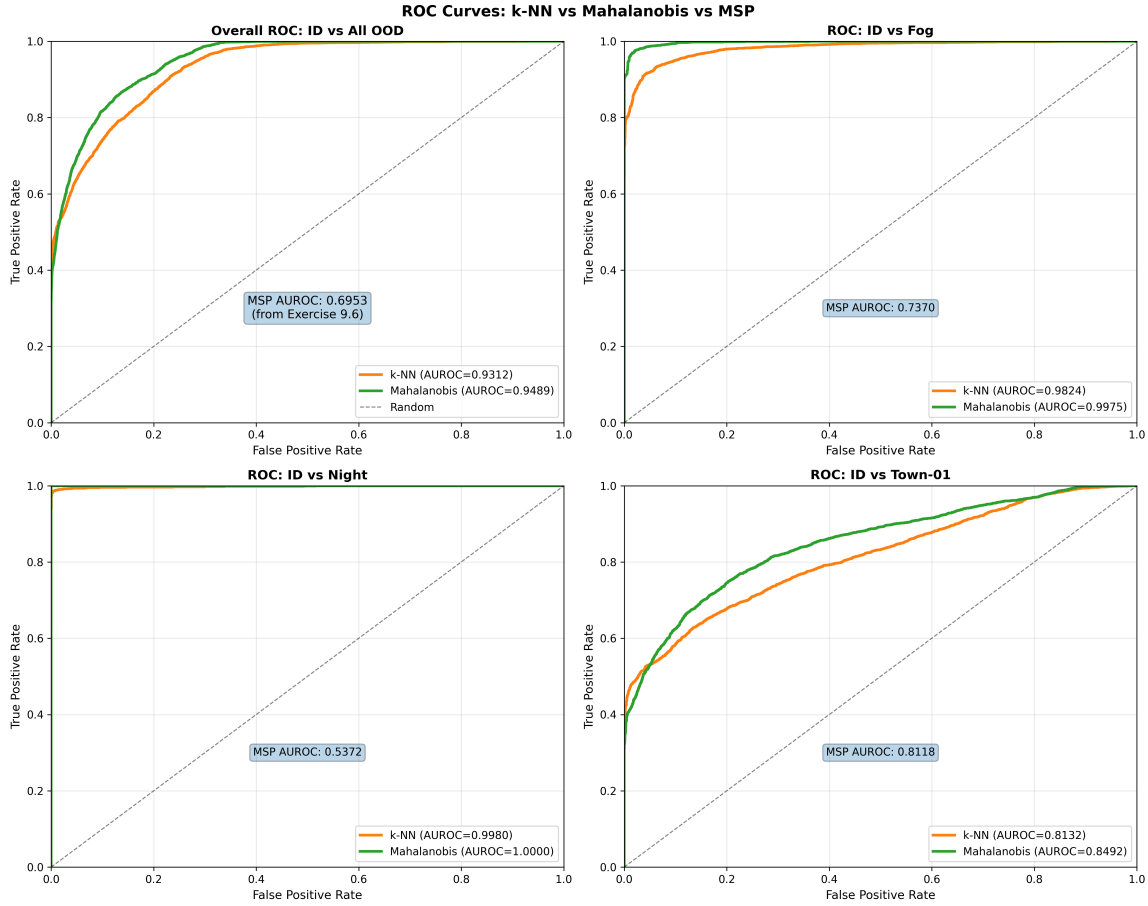


Figure 6: **ROC curve comparison** across OOD detection methods – MSP, k -NN, and Mahalanobis distance – on identical ResNet-18 features (**Exercise 7.7**). Supports the **V-4** verdict rationale that the shortfall is specific to MSP, not the underlying features. (The panel’s aggregate MSP AUROC of 0.6953 is a combined-splits figure, distinct from the per-split values V-4’s verdict uses.)

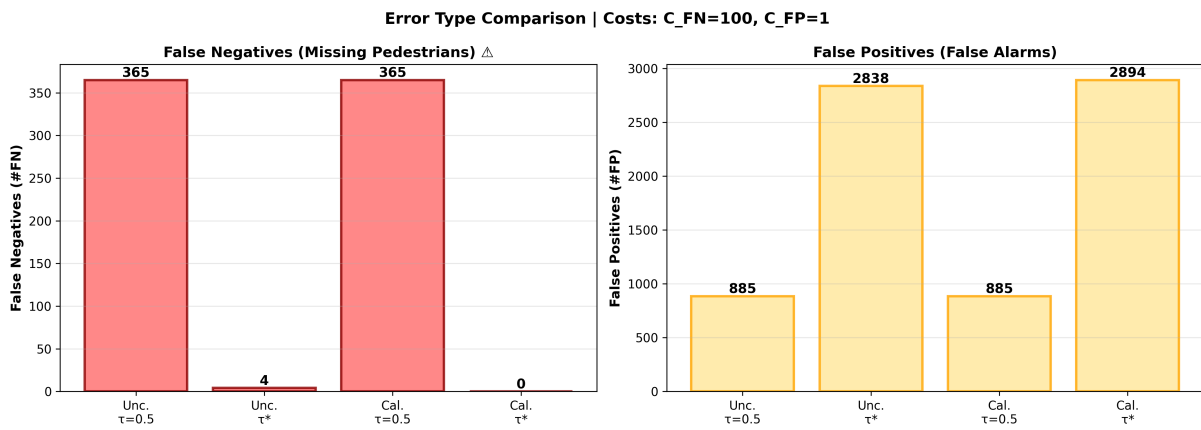


Figure 7: **False negatives vs. false positives** across the standard ($\tau = 0.5$) and cost-optimal ($\tau^* \approx 0.0099$) decision thresholds for the pedestrian detector (**Exercise 9.6**). Visually confirms the **V-5** evidence: false negatives drop from 365 to 0, at the cost of false positives rising from 885 to 2,894.

Scope and Structural Limitations of This Safety Case :

This report’s constraint set expanded organically: Exercise 2 established 8 base constraints (**SC-1–SC-8**), while Exercises 7 and 8 each added a model-level and system-level pair (**SC-9/SC-10**) for OOD detection, (**SC-11/SC-12**) for adversarial robustness), bringing the total to 12. The five fixed verifications (**V-1–V-5**), scoped earlier in the course, do not cover all **12**. This note makes that asymmetry explicit.

1. **Five constraints have no dedicated verification.** V-1–V-5 evidence 7 of 12 constraints (SC-1, SC-2, SC-3, SC-9, SC-10, SC-11, SC-12). SC-4–SC-8 – planner-logic and false-positive constraints – have no exercise-based test and are marked “—” in Section 3.6 rather than assigned a fabricated verdict.
2. **SC-1 and SC-2 carry two verdicts by design, not contradiction.** V-1’s recall evidence finds them **Not met** (pedestrian recall 0.483 vs. ≥ 0.90); V-3’s calibration evidence finds them **Met** via LS-3. These verify different sub-claims – V-1 asks whether the model detects enough pedestrians (no); V-3 asks whether it fails honestly when wrong (yes). Both hold simultaneously without conflict.
3. **Deployment conditions do not imply full coverage.** Even if V-1, V-2, and V-4 were all met, SC-4–SC-8 would remain unverified. Section 6’s restricted-deployment recommendation closes only the failure modes this report actually evidenced, not the full constraint set.
4. **SC-9 anticipates its own fix without asserting it is applied.** SC-9 requires detector choice be “justified by measured performance.” V-4 tested only MSP (AUROC 0.537–0.812, below 0.90); Exercise 7.7 separately measured Mahalanobis and k -NN on the same features as substantially stronger (0.949, 0.931) – cited in the Residual Risks mitigation, but not yet the deployed method.